

Kunj Rathod

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Education

University of Utah

B.S. in Computer Science

Aug 2023 – Dec 2026

GPA: 3.7/4.0, Dean's List

Coursework: Distributed Systems, Operating Systems, Machine Learning, Computer Vision, NLP, Algorithms, Databases

Research

MEMROT v2 — Mechanistic Anatomy of Long-Context Memory Failure

May 2026 – Present

Independent research artifact; Gemma-2-2B, Gemma Scope SAEs, LongMemEval-S, CHPC/SLURM

- Built a reproducible mechanistic-evaluation pipeline for controlled haystack retrieval at fill levels $k \in \{0, 1, 2, 4, 8, 14\}$: gold-position randomization, in-hook attention reduction, empirical SAE layer scans, mean-ablation/intervention tests, and an 840-pass sweep over 140 exactly-one-gold questions.
- Wrote the paper as a single-model case study with explicit limitations around feature selection, attention normalization, sliding-window visibility, shared trends, and heuristic grading; prioritized honest failure analysis over inflated claims.

Circuit-Level Generalization in LLMs via Sparse Feature Circuits

Jan 2026 – Present

University of Utah; attribution graphs, sparse feature circuits, graph skeletonization

- Building automated pipelines to extract sparse feature circuits from default tasks and predict transfer to non-default tasks using attribution graphs and Pathways Discovery; analyzing circuit stability across model families.
- Open-sourced a graph-skeletonization tool that prunes attribution graphs and groups remaining nodes into supernodes via Mapper embeddings or local LLM grouping, exporting visualization-compatible qParams without dropping source metadata.

Disentangling Correct Refusal from Over-Refusal via Sparse Feature Interventions

Mar 2026 – May 2026

Qwen2.5-0.5B-Instruct; sparse autoencoders; refusal-safety evaluation

- Built a 400-example unsafe/benign-overlap benchmark, extracted residual activations, trained sparse autoencoders, ranked refusal-feature families, and tested causal steering/suppression under held-out, paraphrase, and specificity checks.
- Found a selective positive result — amplifying one feature family improved held-out unsafe refusal from 0.80 to 0.85 — plus a useful negative result: suppressing top over-refusal features did not reliably improve benign compliance.

Industry Experience

Software Engineering Intern — Microsoft Fabric (Azure Data)

May 2026 – Aug 2026

Microsoft, Redmond, WA

- Building SKU Change History end-to-end for the Fabric Manage Capacities admin portal: query builder, service layer, REST API, telemetry integration, and auditable SKU-change timelines for enterprise capacity administrators.
- Defining the data contract with UX and backend teams across 3 time zones; using Microsoft's internal AI stack to accelerate debugging, implementation planning, and service-delivery workflows.

Software Development Intern, AI Services (SUDO Program)

Jan 2025 – Present

University of Utah Health, Salt Lake City, UT

- Architected HIPAA-compliant LLM serving infrastructure for 90+ hospital executives: token-streaming inference at p95 <200ms TTFT and 40% latency reduction via pipeline optimization and API caching.
- Owned API design, schema, and AWS infrastructure (Lambda, API Gateway, RDS, DynamoDB, S3, Bedrock); shipped 6 features across 4 sprints and implemented distributed session persistence for 1,000+ concurrent conversations.

AI Engineering Intern

Nov 2024 – Apr 2025

CourtEasy.ai / Nugen (Remote)

- Scaled hybrid retrieval (dense vectors + BM25 + reranking) over 10M+ legal documents, improving recall 28% and reducing hallucinations 35%; benchmarked 8 LLM families on F1, latency, and cost and built model routing that cut inference spend.

Selected Projects

Continuum — Local-First Multimodal Memory Engine

Jun 2026

Rust, Tauri 2, React, TypeScript, LanceDB, OCR, embeddings, MCP

- Built a privacy-first desktop memory layer that captures foreground context on-device, normalizes text/app/window/URL metadata, stores vector + lexical retrieval fields in LanceDB, and exposes agent-ready recall through an MCP server.

HirePilot — Autonomous AI Recruiting Pipeline (Podium AI Hackathon Winner)

Mar 2026

Python/FastAPI, Anthropic, Perplexity, GitHub REST, Twilio, Slack, Resend, Google Calendar

- Shipped a real-integration 9-agent recruiting loop in 24 hours: Slack/API intake, GitHub sourcing, Perplexity enrichment, Anthropic ranking/outreach/interviews, Twilio live voice, Calendar scheduling, and manager approval controls.

Languages & Technologies

Languages: Python, C++, Java, Rust, Swift, TypeScript, SQL

AI/ML: PyTorch, TensorFlow, JAX, sparse autoencoders, attribution graphs, mechanistic interpretability, RAG/GraphRAG, agent evaluation, LangChain, LlamaIndex

Systems & Cloud: AWS (Lambda, API Gateway, S3, DynamoDB, RDS, Bedrock), Azure, Docker, Kubernetes, FastAPI, gRPC, CI/CD, SLURM, distributed systems